

```

39     users: { farmers: 0, agents: 0, total: 0 },
40     listings: { open: 0, delivered: 0 },
41   };
42 }
43 }

```

Listing 8.13: Admin Stats (firebase-service.ts – getAdminStats)

The function uses `getCountFromServer` for efficient aggregation without downloading individual documents, reducing bandwidth consumption. Error handling returns zeroed-out stats to prevent the admin dashboard from crashing if Firestore is temporarily unavailable.

8.13 Waste Delivery Tracking

The waste delivery tracking functions manage the logistics of waste transportation from collection agents to recycling facilities:

```

1  export async function recordWasteDelivery(
2    data: any
3  ): Promise<string> {
4    const payload = {
5      ...data,
6      createdAt: serverTimestamp(),
7      status: 'DELIVERED',
8    };
9    const ref = await addDoc(
10      collection(db, 'wasteDeliveries'), payload
11    );
12
13    if (data.reportId) {
14      const reportRef = doc(db, 'wasteReports', data.reportId);
15      await updateDoc(reportRef, {
16        status: 'IN_TRANSIT',
17        collectionAgentId: data.agentId,
18        collectionAgent: data.agentName,
19        lastUpdate: serverTimestamp(),
20      });
21    }
22    return ref.id;
23  }
24
25  export async function recordWasteProcessing(
26    data: any
27  ): Promise<string> {

```